Writing & Interpreting Linear Models

WORKSHEET | PART 1 OF 3 Questions 1–4

Show your reasoning and label any units.

Question 1 **R&P**

A membership costs $5 per month plus $2 for each movie rented. Write a model for total monthly cost C for n movies. Interpret m and b.

Question 2 **R&P**

The table has points (0,15), (1,13), (2,11), (3,9). Write y = mx + b and interpret its trend.

Question 3 **R&P**

Write the equation of a line through (0,4) and (2,10). Show work for m and b.

Question 4 **R&P**

Choose which rule matches outputs 4, 7, 10 at inputs 0, 1, 2: (a) y=3x+4 (b) y=4x+3 (c) y=7x+4. Explain.

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WORKSHEET | PART 2 OF 3 Questions 5–8

Show your reasoning and label any units.

Question 5 **R&P**

Position is modeled by p = 20 - 5t (meters, seconds). Interpret 20 and -5; predict position at t = 3.

Question 6 **R&P**

Spiral + model check: Is {(0,7),(1,9),(2,11),(2,11)} a function? What linear rule fits these pairs?

Question 7 **Matching practice**

A club charges a $7 monthly fee plus $3 per class. Write cost C for n classes and interpret the rate and starting fee.

Question 8 **Matching practice**

The table contains (0,12), (1,16), (2,20). Write the equation and describe its rate and starting value.

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WORKSHEET | PART 3 OF 3 Questions 9–12

Show your reasoning and label any units.

Question 9 **Matching practice**

Find the equation of the line through (0,6) and (3,0). Show m and b.

Question 10 **Matching practice**

Which rule fits outputs 7, 11, 15 for inputs 0, 1, 2: (a) y = 4x + 7, (b) y = 7x + 4, (c) y = 3x + 7?

Question 11 **Matching practice**

Position follows p = 30 - 4t (meters and seconds). Interpret the rate and find p when t = 5.

Question 12 **Matching practice**

Is {(0,-1), (1,1), (2,3), (2,3)} a function? Write a linear rule that fits.